

ECONOMICS 114: TUTORIAL 4

(Presented in the week of 23 March 2026)

NOTE: Students are expected to work out the tutorial questions as thoroughly as possible before the tutorial class and to show their answers to the tutor during the tutorial class.

Coverage: CORE Unit 4

Question 1 (4 marks)

Indicate whether each of the following statements is True or False and provide a clear justification or explanation for your answer if it is False:

1.1 The tragedy of the commons refers to a situation where individuals work together to preserve and sustainably manage a shared resource.

1.2 In economics, free riding refers to a situation where people enjoy the benefits of a resource, service, or public good without contributing to its cost or maintenance.

Question 2 (4 marks)

2.1 Russel and Alan are two first-year students in an economics class at Stellenbosch University. They both had economics in high school and are under the impression that the upcoming Assessment 1 (A1) will be “easy peasy”. Two weeks ahead of the A1, their lecturer informs them that the pass rates for A1 for recent years are only 80%, meaning that out of 2000 students, 400 on average failed. Both Russel and Alan must decide whether to study hard or procrastinate. The decision by each student is made independently, and without consulting the other. The interaction is represented below.

		ALAN	
		Procrastinate	Study hard
RUSSEL	Procrastinate	8, 8	14, 4
	Study hard	4, 14	12, 10

Which of the following statements regarding the game above is most CORRECT?

- i) (Procrastinate, Procrastinate) is a dominant strategy equilibrium for Russel and Alan.
 - ii) (Study hard, Study hard) is a Nash equilibrium in this game.
 - iii) This game is an example of a prisoner’s dilemma.
 - iv) This game is an example of an invisible hand game.
- a. i only

- b. i and iii
- c. ii and iii
- d. iv only

2.2 Read the following statements about the outcome of an economic interaction and choose the correct option(s).

- a. If the allocation is Pareto efficient, then you cannot make anyone better off without making someone worse off.
- b. If an allocation is Pareto-efficient, it is also fair.
- c. There cannot be more than one Pareto-efficient outcome.
- d. If there are multiple Pareto-efficient outcomes, and an allocation is Pareto-dominated by one Pareto-efficient outcome, then it will also be Pareto-dominated by the other Pareto-efficient outcomes.

Question 3 (3 marks)

Provide the definitions of the following:

3.1 Best response

3.2 Dominant strategy

3.3 Nash equilibrium

Question 4 (6 marks)

Consider a game where two hunters must decide whether to hunt a stag or a hare. The only way to successfully hunt a stag is if both hunters hunt the stag. Thus, when one player hunts a stag and the other a hare, the one who hunted the stag will get nothing.

		Player 2	
		Stag	Hare
Player 1	Stag	3, 3	0, 4
	Hare	4, 0	2, 2

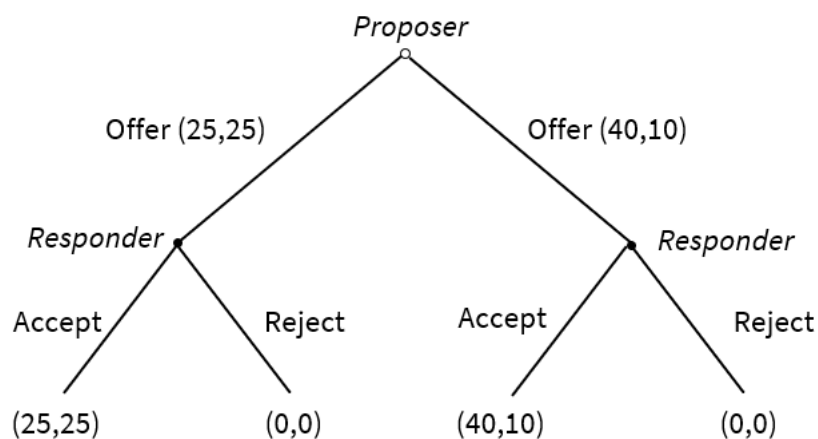
4.1 What is the Nash Equilibrium of this game?

4.2 Draw the four allocations onto a payoff graph with Player One on the x-axis and Player Two on the y-axis.

4.3 What can you conclude about the Pareto efficiency of each of the allocations?

Question 5 (3 marks)

In a lab experiment, behavioural economists want to test the ultimatum game on South African students. By observing their choices, the economists investigate the subjects' preferences and motives, such as pure self-interest, altruism, inequality aversion, or reciprocity. Random pairs of players are assigned to the role of a Proposer and a Responder. The Proposer is given R50 and instructed to offer the Responder part of it. After observing the offer, the Responder can either accept or reject it. If rejected, both get nothing; if accepted, the Proposer gets x , and the Responder gets y .



What is the solution to the above game tree of a simplified ultimatum game?